

# Week 4: Shapes and Ray-Shape Intersections

Intersecting Spheres and Triangles

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CS 4212: Computer Graphics

Spring 2026

## Implement Ray Intersection with Shapes

Send rays into the scene and check if shapes have been intersected or not

- Tue - implicit equations, ray-intersection with the sphere
- Wed - implement and test the ray-sphere intersection
- Thu - verify ray-sphere intersection works, ray-triangle intersection & barycentric coordinates

## Potential issues with ray direction shading

- Our framebuffer image origin is in bottom left; PNG is in upper right with Y going down.
- Does the position of the camera matter for ray direction shading?
- What happens if you want to look straight up or straight down?

Discuss with neighbors and come up with solutions.

# Ray Intersection Tests

Why do we compute an intersection between rays and objects

- Determine if ray leaving the camera hits anything
- If it does, then we can set the color of the pixel based on object hit

We will focus on two types of objects initially

1. Sphere (today)
2. Triangle (Thursday)

Other object types are possible, but any objects that will be rendered **must** support answering an intersection with ray query:

```
bool hit = shape->intersect( ray );
```

We will first focus on ray-sphere intersection to understand how this will work

## Parametric Line Equation (our ray)

$$\vec{p}(t) = \vec{e} + t\vec{d}$$

- $\vec{e}$ : Origin (Camera)
- $\vec{d}$ : Direction
- $t$ : Distance parameter

We need to know IF there is a  $p(t)$  that exists in the scene!

Answering this question and understanding it is the basis for the intersection computation in a ray tracer.

# Implicit Equations

Implicit Equations are ideal for helping us answer if there exists a  $p(t)$  in the scene for a given ray.

What characterizes an *implicit equation*?

- Defines a surface as the set of points  $\vec{p}$  satisfying  $f(\vec{p}) = 0$ .
- Useful for telling us if a point is **on**, **inside**, or **outside** a shape.

For a function  $f(\vec{p})$  describing a solid object (like a sphere):

$f(\vec{p}) = 0 \implies$  Point  $\vec{p}$  is **on the surface**

$f(\vec{p}) < 0 \implies$  Point  $\vec{p}$  is **inside**

$f(\vec{p}) > 0 \implies$  Point  $\vec{p}$  is **outside**

## Ray Intersection Tests

All intersections are between a *parametric line* representation for our ray and typically, an *implicit equation* that describes our shape.

What's an *implicit equation*?

- A function of some number of parameters that when equal to 0, the parameters define the surface (or geometry)
- More formally,  $f(x) = 0$ ,  $f(x, y) = 0$ , or  $f(x, y, z) = 0$  all represent implicit functions
- For instance, consider the implicit line equation for a line:

$$f(x, y) = (y_a - y_b)x + (x_b - x_a)y + x_a y_b - x_b y_a = 0$$

## Implicit Equations, cont'd

Example: *Implicit equation* of a line (this will be useful later in the week too):

$$f(x, y) = (y_a - y_b)x + (x_b - x_a)y + x_a y_b - x_b y_a = 0$$

Consider the line formed between two points  $\vec{a} = (1, 1)$  and  $\vec{b} = (3, 3)$ :

$$f(x, y) = (1 - 3)x + (3 - 1)y + 1 * 3 - 3 * 1 = -2x + 2y$$

Verify that's correct with your neighbor.

- What is  $f(0, 0)$ ?

## Implicit Equations, cont'd

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- What is  $f(0, 0)$ ?
  - $f(0, 0) = 0$  *On the line or not?*
- What is  $f(2, 2)$ ?

## Implicit Equations, cont'd

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  - $f(0, 0) = 0$  *On the line or not?*
- What is  $f(2, 2)$ ?
  - $f(2, 2) = 0$  *On the line or not?*
- What is  $f(1, 2)$ ?

# Implicit Equations, cont'd

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  - $f(0, 0) = 0$  *On the line or not?*
- What is  $f(2, 2)$ ?
  - $f(2, 2) = 0$  *On the line or not?*
- What is  $f(1, 2)$ ?
  - $f(1, 2) = -2 + 4 = 2$  *On the line or not?*
- What is  $f(2, 1)$ ?

# Implicit Equations, cont'd

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  - $f(2, 2) = 0$  *On the line or not?*
- What is  $f(1, 2)$ ?
  - $f(1, 2) = -2 + 4 = 2$  *On the line or not?*
- What is  $f(2, 1)$ ?
  - $f(2, 1) = -4 + 2 = -2$  *On the line or not?*

# Implicit Sphere Equation

The same ideas work for 3D shapes, like spheres:

- Implicit Sphere:  $f(x, y, z) = 0$  when a point  $\vec{p} = (x, y, z)$  is on the sphere
- if  $f(x, y, z) > 0$  or  $f(x, y, z) < 0$  then point  $\vec{p}$  is inside or outside the sphere

Implicit Circle Example (2D, so not a sphere)

$$f(x, y) = x^2 + y^2 - R^2 = 0$$

Sphere would be

$$f(x, y, z) = x^2 + y^2 + z^2 - R^2 = 0$$

## Exercise 1: Implicit Equation Understanding

### Group Task (2 mins)

Given a circle equation:  $f(x, y) = x^2 + y^2 - 4 = 0$  (Radius=2).

Calculate the value of  $f(x, y)$  for these points and classify them (Inside? Outside? On Surface?).

1.  $P_1 = (0, 0)$
2.  $P_2 = (2, 0)$
3.  $P_3 = (3, 4)$

## Exercise 1: Implicit Equation Understanding

### Group Task (2 mins)

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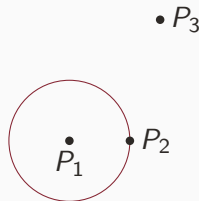
Calculate the value of  $f(x, y)$  for these points and classify them (Inside? Outside? On Surface?).

1.  $P_1 = (0, 0)$

2.  $P_2 = (2, 0)$

3.  $P_3 = (3, 4)$

- $P_1 : 0^2 + 0^2 - 4 = -4 \implies \text{Inside } (< 0)$
- $P_2 : 2^2 + 0^2 - 4 = 0 \implies \text{On Surface } (= 0)$
- $P_3 : 3^2 + 4^2 - 4 = 21 \implies \text{Outside } (> 0)$



# Implicit Sphere Equation

## Implicit Sphere Equation

$$(x_p - x_{center})^2 + (y_p - y_{center})^2 + (z_p - z_{center})^2 - R^2 = 0$$

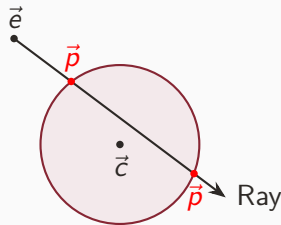
We can transform this into a *vector* equation:

$$(\vec{p} - \vec{center}) \cdot (\vec{p} - \vec{center}) - R^2 = 0$$

So, any point  $\vec{p}$  that satisfies this *is on the sphere*!

- Now, just plug in our parametric line equation for  $\vec{p}$ !
- Recall:  $p(t) = \vec{r}_o + t\vec{r}_d$

$$(\vec{r}_o + t\vec{r}_d - \vec{center}) \cdot (\vec{r}_o + t\vec{r}_d - \vec{center}) - R^2 = 0$$



## Exercise: Define the Function

**Goal:** Write the implicit equation for a specific shape.

We want a sphere with:

- Center  $\vec{c} = (1, 2, 3)$
- Radius  $R = 4$

**Task:**

1. Write the mathematical equation for points  $(x, y, z)$  on this sphere.
2. Convert it to the implicit function form  $f(x, y, z) = \dots$  such that the surface is at 0.

## Exercise: Define the Function

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### Solution

$$(x-1)^2 + (y-2)^2 + (z-3)^2 = 4^2$$

$$f(x, y, z) = (x-1)^2 + (y-2)^2 + (z-3)^2 - 16 = 0$$

# Ray-Sphere Intersection

Putting it together:

- Implicit Sphere:  $f(x, y, z) = 0$  when a point  $\vec{p} = (x, y, z)$  is on the sphere
- if  $(f(x, y, z) > 0$  or  $f(x, y, z) < 0$  then point  $\vec{p}$  is inside or outside the sphere
- We want to find the places where  $f(x, y, z) = 0$ !

Parametric Line Equation for Ray

$$p(\vec{t}) = ray_{origin} + t ray_{dir}$$

We have an intersection with the sphere when

- We have found one or two  $t$  value(s) that provide a specific point(s) along the ray that cut through the sphere

# Ray-Sphere Intersection Strategy

## 1. Ray Equation (Parametric):

$$\vec{p}(t) = \vec{e} + t\vec{d}$$

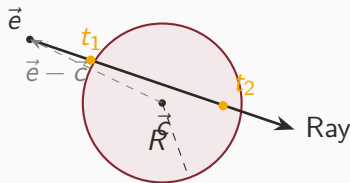
## 2. Sphere Equation (Implicit) (vector-based formulation):

$$(\vec{p} - \vec{c}) \cdot (\vec{p} - \vec{c}) - R^2 = 0$$

## 3. Substitute Ray into Sphere:

$$\underbrace{(\vec{e} + t\vec{d})}_{\vec{p}} - \vec{c} \cdot (\vec{e} + t\vec{d} - \vec{c}) - R^2 = 0$$

This creates an equation where  $t$  is the only unknown!



## Exercise 2: The Setup

### Group Task (3 mins)

We want to find the geometric point where the Ray hits the Sphere.

1. Write down the Ray equation for a point  $\vec{p}(t)$ .
2. Write down the implicit Sphere condition for a point  $\vec{p}$ .
3. **Combine them** into a single equation that must hold true at the intersection.

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### Solution:

$$(\vec{p}(t) - \vec{c}) \cdot (\vec{p}(t) - \vec{c}) - R^2 = 0$$

Substitute  $\vec{p}(t) = \vec{e} + t\vec{d}$ :

### The Intersection Equation

$$(\vec{e} + t\vec{d} - \vec{c}) \cdot (\vec{e} + t\vec{d} - \vec{c}) - R^2 = 0$$

## Exercise 3: The Algebra

### Group Task (5 mins)

Expand the equation to solve for  $t$ .

$$(\underbrace{\vec{e} - \vec{c}}_{\vec{oc}} + t\vec{d}) \cdot (\vec{oc} + t\vec{d}) - R^2 = 0$$

*Hint: Treat  $(\vec{e} - \vec{c})$  as a single vector. Recall that  $(\vec{A} + \vec{B}) \cdot (\vec{A} + \vec{B}) = \vec{A} \cdot \vec{A} + 2\vec{A} \cdot \vec{B} + \vec{B} \cdot \vec{B}$ .*

**Goal:** Get it into the form  $At^2 + Bt + C = 0$ .

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**Goal:** Get it into the form  $At^2 + Bt + C = 0$ .

$$(\vec{d} \cdot \vec{d})t^2 + 2\vec{d} \cdot (\vec{e} - \vec{c})t + (\vec{e} - \vec{c}) \cdot (\vec{e} - \vec{c}) - R^2 = 0$$

- $A = \vec{d} \cdot \vec{d}$
- $B = 2\vec{d} \cdot (\vec{e} - \vec{c})$
- $C = (\vec{e} - \vec{c}) \cdot (\vec{e} - \vec{c}) - R^2$

## Exercise 4: Solving & Interpretation

### Group Task (3 mins)

1. Write the quadratic formula for  $t$ .
2. What part of the formula tells us *how many* intersections there are?
3. Sketch the physical meaning of having 0, 1, or 2 solutions.

## Exercise 4: Solving & Interpretation

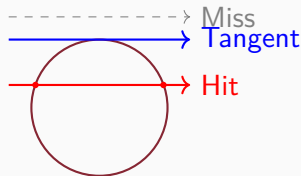
### Group Task (3 mins)

1. Write the quadratic formula for  $t$ . 2. What part of the formula tells us *how many* intersections there are? 3. Sketch the physical meaning of having 0, 1, or 2 solutions.

$$t = \frac{-B \pm \sqrt{B^2 - 4AC}}{2A}$$

**Discriminant:**  $\Delta = B^2 - 4AC$

- $\Delta < 0$ : Miss
- $\Delta = 0$ : Graze (Tangent)
- $\Delta > 0$ : Hit (Enter & Exit)



Expand the dot product:

$$(\vec{d} \cdot \vec{d})t^2 + 2\vec{d} \cdot (\vec{e} - \vec{c})t + (\vec{e} - \vec{c}) \cdot (\vec{e} - \vec{c}) - R^2 = 0$$

into the standard quadratic equation:

$$At^2 + Bt + C = 0$$

### Sketch out the computation

Sketch out pseudo-code what this will turn into with code.

## Solving for $t$

Expand the dot product:

$$(\vec{d} \cdot \vec{d})t^2 + 2\vec{d} \cdot (\vec{e} - \vec{c})t + (\vec{e} - \vec{c}) \cdot (\vec{e} - \vec{c}) - R^2 = 0$$

This is a standard quadratic equation:

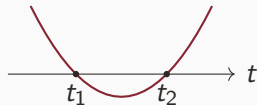
$$At^2 + Bt + C = 0$$

where:

- $A = \vec{d} \cdot \vec{d}$  ( $= 1$  if  $\vec{d}$  normalized)
- $B = 2\vec{d} \cdot (\vec{e} - \vec{c})$
- $C = (\vec{e} - \vec{c}) \cdot (\vec{e} - \vec{c}) - R^2$

Solutions:

$$t = \frac{-B \pm \sqrt{B^2 - 4AC}}{2A}$$



Discriminant  $\Delta = B^2 - 4AC$ :

- $\Delta < 0$ : Miss
- $\Delta = 0$ : Tangent
- $\Delta > 0$ : Two hits

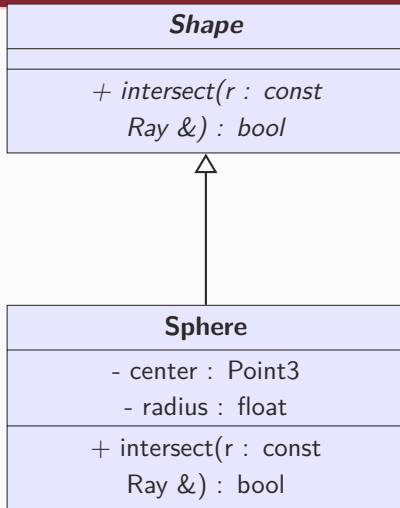
## Lab: Ray-Sphere Intersection

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# What do Shapes need to do?

## Use Abstract Shape class

- Helps to generalize your renderer
- Any object that's a Shape can be rendered
- Currently, only focused on Sphere, but will add more...



```
bool Shape::intersect( const Ray &r ) = 0;
```

## Unit Tests for Sphere Intersection

Create at least three unit tests for your sphere intersection. Create conditions where you will know the resulting answers and can check, such as the following:

- Miss - Verify with condition that does not intersect.
- Two Solutions - Verify with condition that intersects.

Using your Sphere and your ability to generate Perspective Camera rays, can you generate an image of either the Bangladeshi, Japanese, or Palau flags?

## Thursday - More Intersections!

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### More Objects, More Intersections

Render scenes with multiple shapes.

- Modify notions of intersect function to handle multiple shapes
- Add in support for the Triangle

Once you have sphere's and triangles, you can do a lot!

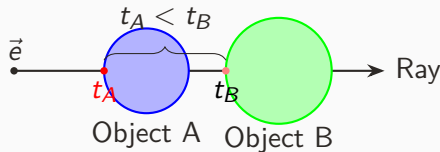
## Multiple Objects: Occlusion and Finding the Closest

In a scene with many objects, a ray might intersect more than one!

- We want the **closest** object that the ray hits.
- "Closest" means the **smallest positive**  $t$  value.

**Remember - This is a Search Algorithm!**

```
 $t_{closest} \leftarrow \infty$   
 $shape_{closest} \leftarrow null$   
for all shapes  $s$  in scene do  
  if  $s \rightarrow intersect(ray, t)$  and  $t < t_{closest}$  then  
     $t_{closest} \leftarrow t$   
     $shape_{closest} \leftarrow s$   
  end if  
end for
```



## Valid Intersections - Behind the Camera?

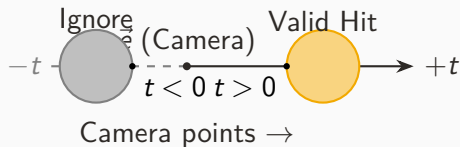
Not all intersections are valid!

- **Behind the ray origin:**  $t < 0$ . These should be ignored.
- **Too close:**  $t < 1.0$  (Inside the “camera” or in front of the image plane).
- **Too far:**  $t > t_{max}$  (Beyond some max value).

**Refined Update:** Only accept specific  $t$  range:

$$t \in [t_{min}, t_{max}]$$

For the camera rays, we will set  $t_{min} = 1.0$ , and initial  $t_{max} = \infty$ .



# What about multiple objects?

## Multiple Objects

What happens when a ray passes through multiple objects?

What's missing from this interface to help us answer this?

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bool Shape::intersect( const Ray &r ) = 0;
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bool Shape::intersect( const Ray &r, const float tmin,  
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## Multiple Objects

What happens when a ray passes through multiple objects?

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bool Shape::intersect( const Ray &r ) = 0;
```

```
bool Shape::intersect( const Ray &r, const float tmin,  
                      float &tmax ) = 0;
```

```
bool Shape::intersect( const Ray &r, const float tmin, float &tmax,  
                      HitStruct &hit ) = 0;
```

Discuss what this function can do for us and what could HitStruct contain?

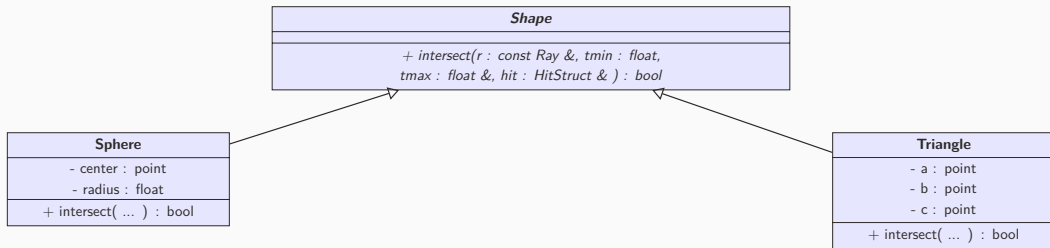
## Multiple Objects

When a ray intersects a shape, we need to get information from it for later.

What's missing from this interface to help us answer this?

```
struct HitStruct {  
};
```

## Adding more Shapes - Triangle



What about triangles? First, need mechanism to determine if point is *in* a triangle!

- Triangles have 3 vertices that lie in the same plane
- Algorithm will use barycentric coordinate representation of a triangle
- Barycentric coordinates form a non-orthogonal coordinate system using two of the edges of the triangle to define a point relative to the triangle's plane
- For instance, a point  $p$  on a triangle defined by three vertices,  $\vec{a}, \vec{b}, \vec{c}$ :

$$\vec{p} = \vec{a} + \beta(\vec{b} - \vec{a}) + \gamma(\vec{c} - \vec{a})$$

# Barycentric Coordinate Systems

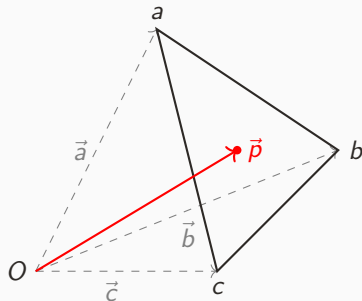
$$\vec{p} = \vec{a} + \beta(\vec{b} - \vec{a}) + \gamma(\vec{c} - \vec{a})$$

and re-working that equation, we get

$$\vec{p} = (1 - \beta - \gamma)\vec{a} + \beta\vec{b} + \gamma\vec{c}$$

let  $\alpha = 1 - \beta - \gamma$ , which also produces the constraint that  $\alpha + \beta + \gamma = 1$

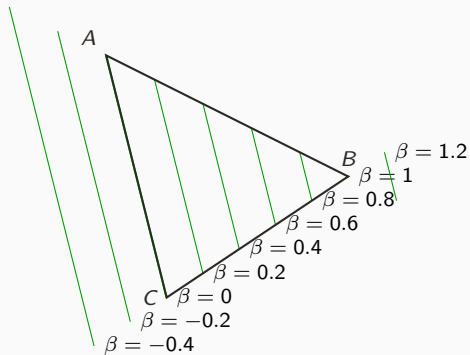
$$\vec{p}(\alpha, \beta, \gamma) = \alpha\vec{a} + \beta\vec{b} + \gamma\vec{c}$$



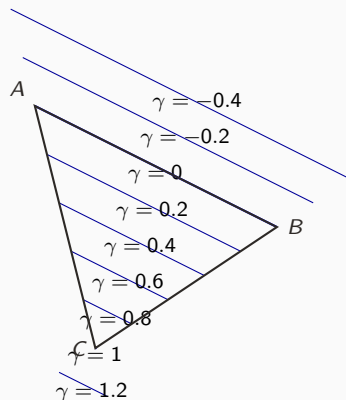
$\vec{p}$  is a weighted average  
of vectors  $\vec{a}$ ,  $\vec{b}$ ,  $\vec{c}$

## Barycentric Grids - Beta Iso-lines

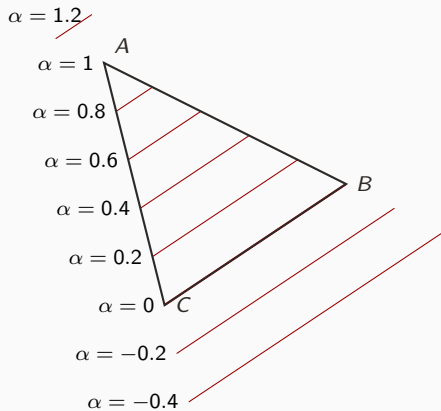
- **Beta** ( $\beta$ ) represents the relative distance from the edge opposite vertex  $B$  (edge  $AC$ ).
- Lines of constant  $\beta$  are parallel to  $AC$ .
- $\beta = 0$  is the line through  $AC$ .
- $\beta = 1$  is the line through  $B$  parallel to  $AC$ .



- **Gamma** ( $\gamma$ ) represents the relative distance from the edge opposite vertex  $C$  (edge  $AB$ ).
- Lines of constant  $\gamma$  are parallel to  $AB$ .
- $\gamma = 0$  is the line through  $AB$ .
- $\gamma = 1$  is the line through  $C$ .



- **Alpha ( $\alpha$ )** is the remaining weight:  $\alpha = 1 - \beta - \gamma$ .
- Corresponds to distance from edge  $BC$ .
- Lines of constant  $\alpha$  are parallel to  $BC$ .

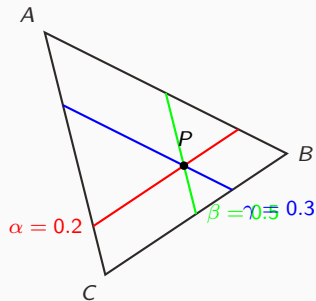


# Barycentric Grids - Combined

Any point  $P$  in the plane is the intersection of these three coordinate lines.

$$P = (0.2, 0.5, 0.3)$$

$$\alpha = 0.2, \beta = 0.5, \gamma = 0.3$$



As far as triangles are concerned, barycentric coordinates have a nice property that a point  $\vec{p}$  is *inside* a triangle **iff**:

$$0 < \alpha < 1$$

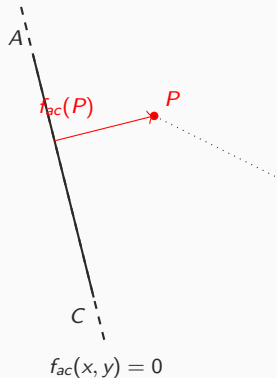
$$0 < \beta < 1$$

$$0 < \gamma < 1$$

*For future note, these coefficients can be used to smoothly interpolate across the surface of a triangle.*

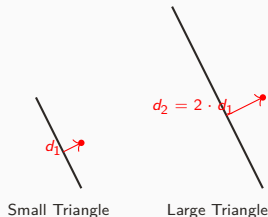
# Implicit Line Equations

- Recall the implicit line equation for the line passing through  $A$  and  $C$ :  $f_{ac}(x, y) = 0$ .
- For a point  $P$  *not* on the line,  $f_{ac}(P)$  returns a value proportional to the signed distance from the line.
- $f_{ac}(P) = d \cdot \|\nabla f\|$ .
- Can we use this "distance" as our coordinate?



# Why Normalize?

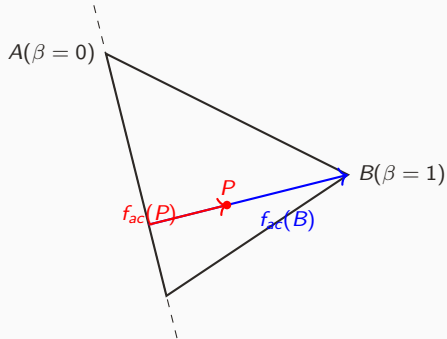
- The raw value  $f_{ac}(P)$  depends on the scale of the triangle!
- If we scale the triangle by 2, the "distance" doubles.
- But the barycentric coordinate  $\beta$  should remain constant for the same *relative* position.
- We need a value that is invariant to scale.



# Deriving Beta

- We define  $\beta$  such that:
  - $\beta = 0$  on the line  $AC$ .
  - $\beta = 1$  at the vertex  $B$ .
- $f_{ac}(B)$  gives the specific "height" of vertex  $B$  from the line  $AC$ .
- By dividing  $f_{ac}(P)$  by  $f_{ac}(B)$ , we get a ratio between 0 and 1.

$$\beta = \frac{f_{ac}(P)}{f_{ac}(B)}$$



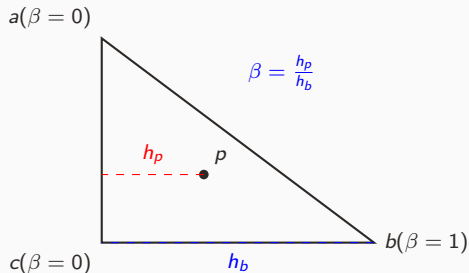
# Barycentric Coordinates Defined

- $\beta$  and  $\gamma$  are easily calculated using the implicit line equation
- Barycentric coordinates represent the signed, scaled distance from the lines on the triangles sides

For instance, we can map  $\beta$  between 0 and 1 across the face of the triangle.

- We would like points on the line  $\vec{a} \rightarrow \vec{c}$  to be where  $\beta = 0$
- Similarly, at point  $\vec{b}$ , we would like  $\beta = 1$

$$\beta = \frac{f_{ac}(x, y)}{f_{ac}(x_b, y_b)}$$



# Barycentric Coordinates Defined

Recall the implicit line equation:

$$f_{ab}(x, y) = (y_a - y_b)x + (x_b - x_a)y + x_a y_b - x_b y_a = 0$$

which when applied to beta let's us normalize our values into the 0 to 1 range for points on the triangle

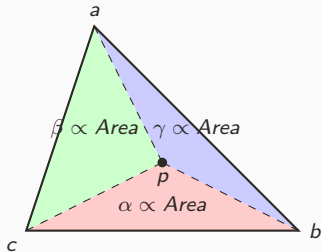
$$\beta = \frac{f_{ac}(x, y)}{f_{ac}(x_b, y_b)}$$

Similarly,  $\gamma$  can be defined:

$$\gamma = \frac{f_{ab}(x, y)}{f_{ab}(x_c, y_c)}$$

with

$$\alpha = 1 - \beta - \gamma$$



# Ray Triangle Intersection

We want the point  $\vec{p}$  to be the point that our ray intersects the plane of the triangle!

Barycentric coordinate values  $\beta$  and  $\gamma$  can tell us if that point is in the triangle by the following checks:

- iff  $\beta > 0$
- iff  $\gamma > 0$
- iff  $\beta + \gamma < 1$

So, set our ray equal to  $\vec{p}$ :

$$\vec{r}_o + t\vec{r}_d = \vec{a} + \beta(\vec{b} - \vec{a}) + \gamma(\vec{c} - \vec{a})$$

# Ray Triangle Intersection

We know everything except  $t, \beta, \gamma$ :

$$\vec{r}_o + t\vec{r}_d = \vec{a} + \beta(\vec{b} - \vec{a}) + \gamma(\vec{c} - \vec{a})$$

We can rewrite this *vector* equation into three component-based equations:

$$x_{r_o} + tx_{r_d} = x_a + \beta(x_b - x_a) + \gamma(x_c - x_a)$$

$$y_{r_o} + ty_{r_d} = y_a + \beta(y_b - y_a) + \gamma(y_c - y_a)$$

$$z_{r_o} + tz_{r_d} = z_a + \beta(z_b - z_a) + \gamma(z_c - z_a)$$

Three equations. Three unknowns. We can solve with a linear system solver:

$$Ax = b$$

# Ray Triangle Intersection

We know everything except  $t, \beta, \gamma$ :

$$\begin{bmatrix} x_a - x_b & x_a - x_c & x_{rd} \\ y_a - y_b & y_a - y_c & y_{rd} \\ z_a - z_b & z_a - z_c & z_{rd} \end{bmatrix} \begin{bmatrix} \beta \\ \gamma \\ t \end{bmatrix} = \begin{bmatrix} x_a - x_e \\ y_a - y_e \\ z_a - z_e \end{bmatrix}$$

You can solve this with Cramer's rule using 3x3 matrices and determinants (if you want):

You can represent this decomposition as a linear system using variables  $a$  through  $l$  to represent the computations:

$$\begin{bmatrix} a & d & g \\ b & e & h \\ c & f & i \end{bmatrix} \begin{bmatrix} \beta \\ \gamma \\ t \end{bmatrix} = \begin{bmatrix} j \\ k \\ l \end{bmatrix}$$

# Ray Triangle Intersection

Using Cramer's rule, you can now solve directly for  $t, \beta, \gamma$ :

$$\begin{bmatrix} a & d & g \\ b & e & h \\ c & f & i \end{bmatrix} \begin{bmatrix} \beta \\ \gamma \\ t \end{bmatrix} = \begin{bmatrix} j \\ k \\ l \end{bmatrix}$$

where

$$a = (x_a - x_b), b = (y_a - y_b), c = (z_a - z_b)$$

$$d = (x_a - x_c), e = (y_a - y_c), f = (z_a - z_c)$$

$$g = x_{r_d}, h = y_{r_d}, i = z_{r_d}$$

$$j = (x_a - x_e), k = (y_a - y_e), l = (z_a - z_e)$$

taken from the standard linear system in the previous slide.

# Ray Triangle Intersection

Solve directly for  $t, \beta, \gamma$  using the following:

$$\beta = \frac{j(ei - hf) + k(gf - di) + l(dh - eg)}{M}$$

$$\gamma = \frac{i(ak - jb) + h(jc - al) + g(bl - kc)}{M}$$

$$t = -\frac{f(ak - jb) + e(jc - al) + d(bl - kc)}{M}$$

$$M = a(ei - hf) + b(gf - di) + c(dh - eg)$$

```
{ Compute t }
if (t < tmin) or (t > tmax) then
    return false;
end if
{ Compute γ }
if (γ < 0) or (γ > 1) then
    return false;
end if
{ Compute β }
if (β < 0) or (β > 1 - γ) then
    return false;
end if
return true;
```